

Pulse of Prevention – Heart Disease Data EDA

Project Overview

Exploratory analysis of heart health measurements to examine how patient characteristics and clinical indicators vary between records with and without heart disease. The notebook uses Python libraries to clean, summarize, and visualize the dataset.

Requirements

Python 3.x and Jupyter Notebook.

Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook.

Data Preparation

Started with 1,025 rows and removed exact duplicate records, leaving 302 unique records. Checked missing values (none reported) and retained the dataset's 14 columns, including the target variable.

Analysis Approach

Reviewed patient age and gender, resting blood pressure, cholesterol, fasting blood sugar, chest pain types, maximum heart rate, exercise-induced angina, ECG results, major vessels, and thalassemia. Compared measurements by heart disease target and explored correlations and combinations of clinical indicators.

Key Findings

The deduplicated dataset contains 302 records: 164 with heart disease and 138 without. Average age is 54.42 years; average resting blood pressure is 131.60 and average cholesterol is 246.50. Chest pain type, maximum heart rate, and exercise-induced angina show notable correlations with the target in this dataset. These are associations, not proof of causation.

Challenges

The source contains repeated records, so deduplication was necessary before analysis. The dataset is a single snapshot per patient and does not support survival or follow-up trend analysis. Logistic regression was outside the notebook's selected library scope and was not included.

Recommendations

Use the observed patterns as a starting point for further investigation, not as a diagnostic tool. Future work could validate findings on a larger independent dataset and apply suitable statistical or predictive methods with careful evaluation.

Conclusion

The project demonstrates data cleaning, exploratory analysis, and visualization of heart health data. It highlights differences and associations in the available measurements while recognizing the limits of a small, snapshot dataset.

Project Files: Heart_eda.ipynb – analysis notebook; heart.csv – source dataset.